

END-TO-END PHYSICAL AI ONCOLOGY CLINICAL TRIAL
UNIFICATION

Adaption: 21 CFR Part 312

Investigational New Drug Application

Modified ECFR

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Prefatory Note

This adaptation extends the prior 21 CFR Part 312 — Investigational New Drug Application to address the end-to-end integration of Physical AI systems, advanced robotics, digital twins, and AI/ML agents into oncology clinical trials. The prior 21 CFR Part 312 regulation established foundational procedures and requirements governing the use of investigational new drugs, including submission, review, and approval of INDs by the Food and Drug Administration. This current document adapts those procedures to encompass the specialized requirements of Physical AI oncology clinical trial unification, including autonomous surgical robots, therapeutic positioning systems, diagnostic needle-placement platforms, rehabilitative exoskeletons, companion monitoring systems, AI-driven decision support, simulation-validated procedures, and federated multi-site trial coordination.

The Unification Standard Level (USL) scoring framework (v1.4.0 through v1.8.0) provides quantitative readiness assessments for robotic platforms across four dimensions: simulation switching, AI integration, cross-robot sharing, and clinical trial collaboration. USL scores range from 1.0 to 10.0 and are referenced throughout this adaptation to establish minimum readiness thresholds for clinical deployment and to inform IND submission requirements for Physical AI components.

The national MCP-PAI standard (v1.2.0) defines the five-server Model Context Protocol topology (authorization, FHIR, DICOM, ledger, provenance), 23 standardized tools, six actor roles, and five cumulative conformance levels that govern data access, audit, and robot agent operations in Physical AI oncology trials. This protocol infrastructure underpins the data management, safety reporting, and electronic records requirements throughout this adaptation.

This adaptation should be read in conjunction with the physical-ai-oncology-trials repository (v2.5.0, DOI: 10.5281/zenodo.19057628), the national MCP-PAI oncology trials standard (v1.2.0, DOI: 10.5281/zenodo.18869776), the ICH E6(R3) Physical AI adaptation (DOI: 10.5281/zenodo.18973368), the 21 CFR Part 50 Physical AI adaptation (DOI: 10.5281/zenodo.19040707), and the federated learning framework (DOI: 10.5281/zenodo.18840880).

Document History

Prior Regulation 21 CFR Part 312 — Investigational New Drug Application. Originally published at 52 FR 8831, March 19, 1987, with subsequent amendments through 2023. The prior regulation established the baseline framework for IND submissions, clinical investigation phases, sponsor and investigator responsibilities, safety reporting, and expanded access provisions.

Current Adaptation End-to-End Physical AI Oncology Clinical Trial Unification: Adaption of 21 CFR Part 312, released 17 March 2026. This adaptation incorporates Physical AI requirements throughout for oncology clinical investigations involving autonomous robotic systems, digital twins, simulation frameworks, and AI/ML agents.

Repository Refs physical-ai-oncology-trials v2.5.0 (March 2026), national-mcp-pai-oncology-trials v1.2.0 (March 2026). These repositories provide validated code, tools, specifications, and documentation supporting this adaptation.

Prior 21 CFR Part 312: Public Domain Notice

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Authority: 21 U.S.C. 321, 331, 351, 352, 353, 355, 360bbb, 371; 42 U.S.C. 262.

Source: 52 FR 8831, Mar. 19, 1987, unless otherwise noted.

Editorial Note: Nomenclature changes to part 312 appear at 69 FR 13717, Mar. 24, 2004.

Change Summary Table

The following table identifies every section of the prior 21 CFR Part 312 regulation that has been modified in this adaptation and summarizes the nature of the Physical AI additions. Sections not listed below retain their original text without modification.

Section	Nature of Physical AI Additions
§ 312.1	Extended scope to include Physical AI systems performing, assisting, or monitoring clinical procedures; added applicability to robotic platforms, simulation frameworks, and digital twins
§ 312.2	Added Physical AI applicability provisions for IND exemptions; clarified that Physical AI system components require IND coverage when used in clinical investigations
§ 312.3	Added 22 Physical AI definitions covering robot agents, USL, digital twins, simulation frameworks, MCP, conformance levels, hash-chained audit trails, federated learning, PCCP, and safety constructs
§ 312.6	Extended labeling requirements to Physical AI system components and robot-administered investigational drugs
§ 312.7	Extended promotion restrictions to Physical AI system capabilities associated with investigational drugs
§ 312.8	Added Physical AI cost recovery provisions for robotic system deployment, simulation validation, and digital twin infrastructure
§ 312.10	Extended waiver provisions to accommodate Physical AI system requirements
§ 312.20	Added IND requirement for Physical AI system components used in clinical investigations
§ 312.21	Added Physical AI considerations for each phase of clinical investigation, including simulation-based Phase 0 validation
§ 312.22	Extended IND submission principles to Physical AI system documentation requirements
§ 312.23	Added comprehensive Physical AI documentation requirements for IND content, including robot specifications, simulation validation data, digital twin models, AI/ML algorithms, and cybersecurity assessments
§ 312.30	Extended protocol amendment requirements to Physical AI system modifications, algorithm updates, and simulation framework changes
§ 312.31	Extended information amendment requirements to Physical AI technical updates
§ 312.32	Added Physical AI safety reporting requirements including robot malfunction events, AI prediction errors, digital twin divergence, and cybersecurity incidents
§ 312.33	Extended annual reporting to Physical AI system performance metrics, USL score updates, and simulation validation results
<i>Continued on next page</i>	

Section	Nature of Physical AI Additions
§ 312.38	Extended IND withdrawal procedures to Physical AI system decommissioning
§ 312.40	Added Physical AI general requirements for IND effectiveness, including simulation validation and USL threshold verification
§ 312.42	Added Physical AI grounds for clinical hold including robotic safety failures, AI model degradation, and simulation-reality divergence
§ 312.44	Extended termination grounds to Physical AI system failures and cybersecurity compromises
§ 312.45	Extended inactive status provisions to Physical AI system dormancy and reactivation
§ 312.47	Added Physical AI system demonstration requirements for regulatory meetings
§ 312.48	Extended dispute resolution to Physical AI technical disagreements
§ 312.50	Extended sponsor responsibilities to Physical AI system selection, validation, monitoring, and lifecycle management
§ 312.52	Extended CRO transfer provisions to Physical AI obligations
§ 312.53	Extended investigator and monitor selection to Physical AI qualifications
§ 312.54	Extended emergency research monitoring to Physical AI systems
§ 312.55	Extended investigator information requirements to Physical AI system updates
§ 312.56	Extended ongoing investigation review to Physical AI system compliance monitoring
§ 312.57	Extended recordkeeping to Physical AI system logs, simulation data, and digital twin records
§ 312.58	Extended inspection access to Physical AI system records and audit trails
§ 312.59	Extended drug disposition to Physical AI system decommissioning procedures
§ 312.60	Extended investigator responsibilities to Physical AI system oversight
§ 312.61	Extended drug control to Physical AI system access control
§ 312.62	Extended investigator recordkeeping to Physical AI case histories and system interaction logs
§ 312.64	Extended investigator reporting to Physical AI system events and performance metrics
§ 312.66	Extended IRB review assurance to Physical AI system components
§ 312.68	Extended inspection access to Physical AI investigator records
§ 312.69	Extended controlled substance handling to Physical AI-mediated administration
§ 312.70	Extended disqualification to Physical AI system misuse and noncompliance
<i>Continued on next page</i>	

Section	Nature of Physical AI Additions
§ 312.80	Extended life-threatening illness provisions to Physical AI accelerated development pathways
§ 312.82	Extended early consultation to Physical AI system design review
§ 312.83	Extended treatment protocols to Physical AI-assisted therapeutic delivery
§ 312.84	Extended risk-benefit analysis to Physical AI system risk profiles
§ 312.85	Extended Phase 4 studies to Physical AI post-market performance monitoring
§ 312.87	Extended active monitoring to Physical AI system clinical performance
§ 312.88	Extended patient safety safeguards to Physical AI-specific protections
§ 312.110	Extended import/export requirements to Physical AI system components and software
§ 312.120	Extended foreign clinical study acceptance to Physical AI system validation data
§ 312.130	Extended public disclosure provisions to Physical AI system data
§ 312.160	Extended laboratory use provisions to Physical AI preclinical testing
§ 312.300	Extended expanded access scope to Physical AI-assisted treatment
§ 312.305	Extended expanded access requirements to Physical AI system safeguards
§ 312.310	Extended individual patient access to Physical AI-assisted emergency treatment
§ 312.315	Extended intermediate-size access to Physical AI multi-site deployment
§ 312.320	Extended treatment IND to Physical AI widespread treatment use
New J	New Subpart J: Additional Requirements for Physical AI Investigational Systems in Oncology Trials

1 SUBPART A — GENERAL PROVISIONS

1.1 § 312.1 Scope

(a) This part contains procedures and requirements governing the use of investigational new drugs, including procedures and requirements for the submission to, and review by, the Food and Drug Administration of investigational new drug applications (IND's). An investigational new drug for which an IND is in effect in accordance with this part is exempt from the premarketing approval requirements that are otherwise applicable and may be shipped lawfully for the purpose of conducting clinical investigations of that drug.

(b) References in this part to regulations in the Code of Federal Regulations are to chapter I of title 21, unless otherwise noted.

1.1.1 Physical AI Adaptation of § 312.1

(c) This part additionally applies to all clinical investigations in which Physical AI systems perform, assist, or monitor procedures involving the administration or dispensing of investigational new drugs to human subjects. Physical AI systems encompass the seven robot categories recognized for clinical investigation purposes: autonomous surgical robots, cobots, humanoid robots, therapeutic robots, diagnostic robots, assistive robots, and rehabilitative robots. These categories are further organized into five functional types defined in the national MCP-PAI standard (v1.2.0): autonomous surgical robots, therapeutic positioning systems, diagnostic needle-placement platforms, rehabilitative exoskeletons, and companion monitoring systems.

(d) The scope of this part extends to IND applications involving investigational drugs that are administered, dispensed, prepared, compounded, or monitored by Physical AI systems. When a Physical AI system participates in any step of the drug administration pathway, the IND must document the system's role, safety parameters, and operational specifications as described in § 312.23 of this part. The processing of investigational drug data through Physical AI systems, including through the five MCP servers (trialmcp-authz, trialmcp-fhir, trialmcp-dicom, trialmcp-ledger, trialmcp-provenance), constitutes an activity subject to the requirements of this part.

(e) The Unification Standard Level (USL) framework serves as the readiness assessment methodology for Physical AI platforms used in clinical investigations under this part. USL scores are computed across four equally weighted dimensions (simulation switching, AI integration, cross-robot sharing, clinical trial collaboration) and range from 1.0 (Minimal) to 10.0 (Reference). The national MCP-PAI standard specifies minimum USL thresholds by procedure type: surgical robots require USL 7.0 or above, therapeutic positioning and diagnostic needle placement require USL 6.0 or above, rehabilitative exoskeletons require USL 4.0 or above, and companion monitoring systems require USL 3.0 or above. These thresholds apply to Physical AI systems used in clinical investigations conducted under INDs submitted pursuant to this part.

(f) Four simulation frameworks are recognized for pre-clinical validation of Physical AI systems before their use in IND-covered clinical investigations: NVIDIA Isaac Lab (v2.3.1) for GPU-accelerated robot training, MuJoCo (v3.4.0) for precision physics simulation, Gazebo Sim (v10.0.0) for ROS 2 integrated simulation, and PyBullet (v3.2.5) for rapid prototyping. Cross-framework validation via the unification bridge must demonstrate behavioral consistency with trajectory deviations below 2mm and force discrepancies below 0.5N before a Physical AI system may be used in clinical investigations under an IND.

(g) Digital twin models used for treatment planning, procedure rehearsal, or real-time intraoperative guidance in connection with investigational drug administration are subject to the documentation and validation requirements of this part. Digital twins must be calibrated to patient-specific imaging data and continuously validated against clinical outcomes throughout the investigation.

[52 FR 8831, Mar. 19, 1987; 46 FR 8979, Jan. 27, 1981. Physical AI adaptation added 17 March 2026.]

1.2 § 312.2 Applicability

(a) *Applicability.* Except as provided in this section, this part applies to all clinical investigations of products that are subject to section 505 of the Federal Food, Drug, and Cosmetic Act or to the licensing provisions of the Public Health Service Act (58 Stat. 632, as amended (42 U.S.C. 201 *et seq.*)).

(b) *Exemptions.*

(1) The clinical investigation of a drug product that is lawfully marketed in the United States is exempt from the requirements of this part if all the following apply:

- (i) The investigation is not intended to be reported to FDA as a well-controlled study in support of a new indication for use nor intended to be used to support any other significant change in the labeling for the drug;
- (ii) If the drug that is undergoing investigation is lawfully marketed as a prescription drug product, the investigation is not intended to support a significant change in the advertising for the product;
- (iii) The investigation does not involve a route of administration or dosage level or use in a patient population or other factor that significantly increases the risks (or decreases the acceptability of the risks) associated with the use of the drug product;
- (iv) The investigation is conducted in compliance with the requirements for institutional review set forth in part 56 and with the requirements for informed consent set forth in part 50; and
- (v) The investigation is conducted in compliance with the requirements of § 312.7.

(2)

- (i) A clinical investigation involving an in vitro diagnostic biological product listed in paragraph (b)(2)(ii) of this section is exempt from the requirements of this part if (a) it is intended to be used in a diagnostic procedure that confirms the diagnosis made by another, medically established, diagnostic product or procedure and (b) it is shipped in compliance with § 312.160.
- (ii) In accordance with paragraph (b)(2)(i) of this section, the following products are exempt from the requirements of this part: (a) blood grouping serum; (b) reagent red blood cells; and (c) anti-human globulin.

(3) A drug intended solely for tests in vitro or in laboratory research animals is exempt from the requirements of this part if shipped in accordance with § 312.160.

- (4) FDA will not accept an application for an investigation that is exempt under the provisions of paragraph (b)(1) of this section.
- (5) A clinical investigation involving use of a placebo is exempt from the requirements of this part if the investigation does not otherwise require submission of an IND.
- (6) A clinical investigation involving an exception from informed consent under § 50.24 of this chapter is not exempt from the requirements of this part.
- (c) *Bioavailability studies.* The applicability of this part to in vivo bioavailability studies in humans is subject to the provisions of § 320.31.
- (d) *Unlabeled indication.* This part does not apply to the use in the practice of medicine for an unlabeled indication of a new drug product approved under part 314 or of a licensed biological product.
- (e) *Guidance.* FDA may, on its own initiative, issue guidance on the applicability of this part to particular investigational uses of drugs. On request, FDA will advise on the applicability of this part to a planned clinical investigation.

1.2.1 Physical AI Adaptation of § 312.2

- (f) *Physical AI system components.* When a clinical investigation involves both an investigational new drug and a Physical AI system that administers, monitors, or otherwise participates in the drug delivery pathway, the IND requirements of this part apply to the combined drug-device-robot system. The exemptions in paragraph (b)(1) of this section do not apply if the introduction of a Physical AI system into the drug administration pathway significantly increases the risks or introduces new categories of risk (including robotic malfunction, AI prediction error, cyber-physical failure, or human-robot interaction hazards) not present in conventional drug administration.
- (g) *Simulation-only investigations.* Investigations conducted entirely in simulation environments (NVIDIA Isaac Lab, MuJoCo, Gazebo Sim, or PyBullet) without any human subject involvement are not subject to the requirements of this part. However, simulation data generated for the purpose of supporting a subsequent IND submission must comply with the documentation standards specified in § 312.23 and must be generated using validated simulation frameworks meeting the specifications in the physical-ai-oncology-trials repository (v2.5.0).
- (h) *Digital twin investigations.* Clinical investigations that use patient-specific digital twin models for treatment planning or procedure rehearsal, where the digital twin output directly influences the administration of an investigational drug to a human subject, are subject to the requirements of this part. Digital twin models used solely for post-hoc analysis or research purposes that do not influence active treatment decisions are not subject to IND requirements but must be documented in the annual report under § 312.33.

[52 FR 8831, Mar. 19, 1987, as amended at 61 FR 51529, Oct. 2, 1996; 64 FR 401, Jan. 5, 1999. Physical AI adaptation added 17 March 2026.]

1.3 § 312.3 Definitions and Interpretations

- (a) The definitions and interpretations of terms contained in section 201 of the Act apply to those terms when used in this part:
- (b) The following definitions of terms also apply to this part:

Act means the Federal Food, Drug, and Cosmetic Act (secs. 201–902, 52 Stat. 1040 *et seq.*, as amended (21 U.S.C. 301–392)).

Clinical investigation means any experiment in which a drug is administered or dispensed to, or used involving, one or more human subjects. For the purposes of this part, an experiment is any use of a drug except for the use of a marketed drug in the course of medical practice.

Contract research organization means a person that assumes, as an independent contractor with the sponsor, one or more of the obligations of a sponsor, e.g., design of a protocol, selection or monitoring of investigations, evaluation of reports, and preparation of materials to be submitted to the Food and Drug Administration.

FDA means the Food and Drug Administration.

IND means an investigational new drug application. For purposes of this part, “IND” is synonymous with “Notice of Claimed Investigational Exemption for a New Drug.”

Independent ethics committee (IEC) means a review panel that is responsible for ensuring the protection of the rights, safety, and well-being of human subjects involved in a clinical investigation and is adequately constituted to provide assurance of that protection. An institutional review board (IRB), as defined in § 56.102(g) of this chapter and subject to the requirements of part 56 of this chapter, is one type of IEC.

Investigational new drug means a new drug or biological drug that is used in a clinical investigation. The term also includes a biological product that is used in vitro for diagnostic purposes. The terms “investigational drug” and “investigational new drug” are deemed to be synonymous for purposes of this part.

Investigator means an individual who actually conducts a clinical investigation (*i.e.*, under whose immediate direction the drug is administered or dispensed to a subject). In the event an investigation is conducted by a team of individuals, the investigator is the responsible leader of the team. “Subinvestigator” includes any other individual member of that team.

Marketing application means an application for a new drug submitted under section 505(b) of the act or a biologics license application for a biological product submitted under the Public Health Service Act.

Sponsor means a person who takes responsibility for and initiates a clinical investigation. The sponsor may be an individual or pharmaceutical company, governmental agency, academic institution, private organization, or other organization. The sponsor does not actually conduct the investigation unless the sponsor is a sponsor-investigator. A person other than an individual that uses one or more of its own employees to conduct an investigation that it has initiated is a sponsor, not a sponsor-investigator, and the employees are investigators.

Sponsor-Investigator means an individual who both initiates and conducts an investigation, and under whose immediate direction the investigational drug is administered or dispensed. The term does not include any person other than an individual. The requirements applicable to a sponsor-investigator under this part include both those applicable to an investigator and a sponsor.

Subject means a human who participates in an investigation, either as a recipient of the investigational new drug or as a control. A subject may be a healthy human or a patient with a disease.

[52 FR 8831, Mar. 19, 1987, as amended at 64 FR 401, Jan. 5, 1999; 64 FR 56449, Oct. 20, 1999; 73 FR 22815, Apr. 28, 2008]

1.3.1 Physical AI Definitions

The following additional definitions apply to clinical investigations involving Physical AI systems. These definitions are derived from the physical-ai-oncology-trials repository (v2.5.0) and the national MCP-PAI oncology trials standard (v1.2.0).

(c) *Physical AI system* means an AI system operating in the physical world through a robotic platform used in clinical investigations involving investigational new drugs. Physical AI systems include autonomous surgical robots, therapeutic positioning systems, diagnostic needle-placement platforms, rehabilitative exoskeletons, and companion monitoring systems. The term encompasses the robotic hardware, its control software, embedded AI/ML models, and associated sensor systems that collectively perform, assist, or monitor clinical procedures involving the administration or dispensing of investigational drugs to human subjects.

(d) *Robot agent* means an autonomous Physical AI system executing clinical procedures within an oncology trial conducted under an IND, interacting through MCP-mediated data access. The robot agent operates under the six-step workflow defined in the national MCP-PAI standard: authenticate, retrieve clinical data, access imaging, execute procedure, record audit, and record provenance. Robot agents are one of six actor types (alongside Trial Coordinator, Data Monitor, Auditor, Sponsor, and CRO) in the MCP actor model.

(e) *Unification Standard Level (USL)* means a scoring framework (v1.4.0 through v1.8.0) evaluating Physical AI platform readiness across four equally weighted dimensions: simulation switching, AI integration, cross-robot sharing, and clinical trial collaboration. USL scores range from 1.0 to 10.0, mapping to ten levels from Minimal (1.0) to Reference (10.0) and three readiness bands: Foundational (1.0 to 4.9), Intermediate (5.0 to 6.9), and Advanced (7.0 to 10.0). Nine robotic platforms have been evaluated: da Vinci dVRK (USL 7.1), Franka Emika Panda (USL 7.4), Boston Dynamics Atlas (USL 5.8), Kinova Gen3 (USL 5.7), Medtronic Hugo RAS (USL 4.5), Agility Digit (USL 4.2), Tesla Optimus (USL 3.6), CMR Versius (USL 3.4), and UFACTORY xArm 7 (USL 3.4).

(f) *Digital twin* means a patient-specific computational model used for treatment simulation, procedure planning, and real-time intraoperative guidance in connection with the administration of an investigational drug. Digital twins are calibrated to longitudinal imaging data using reaction-diffusion equations, pharmacokinetic/pharmacodynamic models, or other mathematical frameworks and are continuously updated with real-world clinical data. Digital twin models used in IND-covered investigations must meet the validation requirements specified in this part.

(g) *Simulation framework* means a physics-based computational environment used for validating Physical AI system behavior before clinical deployment in IND-covered investigations. Recognized simulation frameworks include NVIDIA Isaac Lab (v2.3.1), MuJoCo (v3.4.0), Gazebo Sim (v10.0.0), and PyBullet (v3.2.5). The validated integration path follows: training (Isaac Lab) to validation (MuJoCo) to hardware deployment to clinical use.

(h) *Robot capability profile* means a machine-readable specification of a Physical AI platform's capabilities, safety prerequisites, USL score, and required MCP tools. The profile is defined by a JSON schema (robot-capability-profile.schema.json) in the national MCP-PAI standard and must be registered with the authorization server before any clinical procedure involving an investigational

drug.

(i) *Task order* means a scheduled clinical trial task assigned to a Physical AI system, including procedure type, robot assignment, safety check requirements, and lifecycle state. Task orders progress through defined states: scheduled, safety_check, in_progress, and completed, cancelled, or failed.

(j) *Emergency stop* means an immediate halt capability that must be available for any Physical AI system during clinical procedures involving investigational drugs, with mandatory audit recording. Activation of an emergency stop transitions the task order to a failed state and requires a new safety check cycle before any subsequent procedure.

(k) *Human oversight* means the requirement that qualified clinical personnel maintain supervisory authority over Physical AI systems during all phases of clinical investigation under an IND. Human oversight encompasses the ability to monitor, intervene, override, or terminate Physical AI system operations at any time. The human oversight quality management framework (v1.2.0) specifies risk-tiered controls with escalating requirements based on clinical consequence.

(l) *Hash-chained audit trail* means an immutable, tamper-evident record of all Physical AI system actions using SHA-256 hash chains. Each audit record contains the action performed, actor identity, timestamp, resource affected, and a hash linking it to the previous record. This mechanism satisfies the electronic records requirements of 21 CFR Part 11 for IND-related activities.

(m) *Model Context Protocol (MCP)* means an open protocol for connecting AI agents to external tools and data sources, implemented through five standardized servers: trialmcp-authz (authorization), trialmcp-fhir (clinical data), trialmcp-dicom (imaging), trialmcp-ledger (audit), and trialmcp-provenance (data lineage). The protocol defines 23 tools and operates under the governance of the Linux Foundation AI and Data Foundation.

(n) *Deny-by-default authorization* means a security model where all Physical AI system access to clinical data associated with an IND is denied unless explicitly permitted by policy. Token-based sessions with defined scopes and time limits govern every data access request. This authorization model is enforced by the trialmcp-authz server.

(o) *HIPAA Safe Harbor de-identification* means the removal or generalization of all 18 Protected Health Information (PHI) identifiers before clinical data is provided to Physical AI systems. HMAC-SHA256 with site-specific salts is used for pseudonymization of subject identifiers in IND-related records processed by Physical AI systems.

(p) *Predetermined Change Control Plan (PCCP)* means a documented plan for managing modifications to Physical AI systems used as or within medical devices in connection with investigational drug administration, per FDA AI/ML guidance. The PCCP specifies the types of anticipated modifications, the methodology for implementing and validating changes, and the assessment of the impact on safety and effectiveness.

(q) *Federated learning* means multi-site AI model training without sharing raw patient data, using differential privacy (epsilon and delta parameters) and secure aggregation. The federation framework supports FedAvg, FedProx, and SCAFFOLD algorithms with data harmonization for DICOM, FHIR, ICD-10 to SNOMED CT, and LOINC tumor marker standardization. Federated learning in IND-covered investigations must comply with the data protection requirements of this part.

- (r) *Pre-procedure safety matrix* means the set of verification checks that must pass before a Physical AI system may begin a clinical procedure involving an investigational drug. The matrix includes: valid authorization token, patient identity verification, clinical data availability (FHIR), imaging data access (DICOM), robot readiness and USL threshold verification, environmental checks, simulation validation confirmation, and digital twin synchronization status.
- (s) *Conformance level* means one of five cumulative levels defined in the national MCP-PAI standard for MCP server implementations: Core (base audit and authorization), Clinical Read (FHIR R4 clinical data access), Imaging (DICOM imaging with RECIST), Federated Site (multi-site federated learning with differential privacy), and Robot Procedure (full robot agent workflow with safety matrix and USL integration).
- (t) *Sim-to-real gap* means the difference between the behavior of a Physical AI system in simulation and its behavior in the physical world. For IND-covered investigations, the sim-to-real gap must be quantified using the validation benchmark suite (v1.0) and must demonstrate trajectory deviations below 2mm and force discrepancies below 0.5N for the specific clinical procedures to be performed.
- (u) *Cross-framework validation* means the verification that a Physical AI system's behavior is consistent across multiple simulation frameworks. Cross-framework validation requires testing in at least two recognized simulation frameworks with behavioral consistency metrics within the tolerances specified in the physical-ai-oncology-trials repository (v2.5.0).
- (v) *Vision-language-action (VLA) model* means a generative AI model that accepts visual and language inputs and produces robotic action outputs. VLA models used in Physical AI systems for IND-covered investigations must be validated for the specific clinical procedures to be performed, with performance metrics documented in the IND submission.

[52 FR 8831, Mar. 19, 1987, as amended at 64 FR 401, Jan. 5, 1999; 64 FR 56449, Oct. 20, 1999; 73 FR 22815, Apr. 28, 2008. Physical AI definitions added 17 March 2026.]

1.4 § 312.6 Labeling of an Investigational New Drug

- (a) The immediate package of an investigational new drug intended for human use shall bear a label with the statement “Caution: New Drug—Limited by Federal (or United States) law to investigational use.”
- (b) The label or labeling of an investigational new drug shall not bear any statement that is false or misleading in any particular and shall not represent that the investigational new drug is safe or effective for the purposes for which it is being investigated.
- (c) The appropriate FDA Center Director, according to the procedures set forth in §§ 201.26 or 610.68 of this chapter, may grant an exception or alternative to the provision in paragraph (a) of this section, to the extent that this provision is not explicitly required by statute, for specified lots, batches, or other units of a human drug product that is or will be included in the Strategic National Stockpile.

1.4.1 Physical AI Adaptation of § 312.6

- (d) When an investigational new drug is administered, dispensed, or monitored by a Physical AI system, the labeling of the investigational drug shall additionally include, either on the label or in accompanying documentation provided to the Physical AI system operator:

- (i) A statement identifying the drug as an investigational new drug under an active IND;
 - (ii) The specific Physical AI system(s) authorized to administer, dispense, or monitor the drug, identified by system type, manufacturer, and model;
 - (iii) Any Physical AI system-specific administration parameters, including dosing rate constraints, positional accuracy requirements, and timing specifications that differ from manual administration; and
 - (iv) Machine-readable identifiers (barcode, RFID, or equivalent) sufficient to enable the Physical AI system to verify drug identity, lot number, and expiration date prior to administration.
- (e) The labeling of a Physical AI system component that is integral to the administration of an investigational new drug (such as a robotic injector cartridge, automated infusion adapter, or needle-placement guide) shall bear the statement “Caution: For use with investigational new drug under IND—Physical AI system component.”

[52 FR 8831, Mar. 19, 1987, as amended at 72 FR 73599, Dec. 28, 2007. Physical AI adaptation added 17 March 2026.]

1.5 § 312.7 Promotion of Investigational Drugs

- (a) *Promotion of an investigational new drug.* A sponsor or investigator, or any person acting on behalf of a sponsor or investigator, shall not represent in a promotional context that an investigational new drug is safe or effective for the purposes for which it is under investigation or otherwise promote the drug. This provision is not intended to restrict the full exchange of scientific information concerning the drug, including dissemination of scientific findings in scientific or lay media. Rather, its intent is to restrict promotional claims of safety or effectiveness of the drug for a use for which it is under investigation and to preclude commercialization of the drug before it is approved for commercial distribution.
- (b) *Commercial distribution of an investigational new drug.* A sponsor or investigator shall not commercially distribute or test market an investigational new drug.
- (c) *Prolonging an investigation.* A sponsor shall not unduly prolong an investigation after finding that the results of the investigation appear to establish sufficient data to support a marketing application.

1.5.1 Physical AI Adaptation of § 312.7

- (d) *Physical AI system promotion restrictions.* The promotion restrictions in paragraphs (a) through (c) of this section extend to claims regarding the performance, accuracy, or superiority of Physical AI systems used in connection with the administration of investigational new drugs. A sponsor or investigator shall not represent, in a promotional context, that a Physical AI system’s involvement in drug administration enhances the safety or effectiveness of an investigational drug beyond what the clinical evidence supports.
- (e) *Simulation and digital twin data.* The dissemination of simulation results, digital twin predictions, or AI model performance metrics generated during an IND-covered investigation is permitted as part of the full exchange of scientific information, provided such dissemination does not constitute a promotional claim of drug safety or effectiveness. Simulation data presented in scientific communications shall be clearly labeled as computational predictions subject to validation.

[52 FR 8831, Mar. 19, 1987, as amended at 52 FR 19476, May 22, 1987; 67 FR 9585, Mar. 4, 2002; 74 FR 40899, Aug. 13, 2009. Physical AI adaptation added 17 March 2026.]

1.6 § 312.8 Charging for Investigational Drugs Under an IND

(a) *General criteria for charging.*

- (1) A sponsor must meet the applicable requirements in paragraph (b) of this section for charging in a clinical trial or paragraph (c) of this section for charging for expanded access to an investigational drug for treatment use under subpart I of this part, except that sponsors need not fulfill the requirements in this section to charge for an approved drug obtained from another entity not affiliated with the sponsor for use as part of the clinical trial evaluation (e.g., in a clinical trial of a new use of the approved drug, for use of the approved drug as an active control).
- (2) A sponsor must justify the amount to be charged in accordance with paragraph (d) of this section.
- (3) A sponsor must obtain prior written authorization from FDA to charge for an investigational drug.
- (4) FDA will withdraw authorization to charge if it determines that charging is interfering with the development of a drug for marketing approval or that the criteria for the authorization are no longer being met.

(b) *Charging in a clinical trial.*

- (1) *Charging for a sponsor's drug.* A sponsor who wishes to charge for its investigational drug, including investigational use of its approved drug, must:
 - (i) Provide evidence that the drug has a potential clinical benefit that, if demonstrated in the clinical investigations, would provide a significant advantage over available products in the diagnosis, treatment, mitigation, or prevention of a disease or condition;
 - (ii) Demonstrate that the data to be obtained from the clinical trial would be essential to establishing that the drug is effective or safe for the purpose of obtaining initial approval of a drug, or would support a significant change in the labeling of an approved drug; and
 - (iii) Demonstrate that the clinical trial could not be conducted without charging because the cost of the drug is extraordinary to the sponsor.
- (2) *Duration of charging in a clinical trial.* Unless FDA specifies a shorter period, charging may continue for the length of the clinical trial.

(c) *Charging for expanded access to investigational drug for treatment use.*

- (1) A sponsor who wishes to charge for expanded access to an investigational drug for treatment use under subpart I of this part must provide reasonable assurance that charging will not interfere with developing the drug for marketing approval.
- (2) For expanded access under § 312.320 (treatment IND or treatment protocol), such assurance must include:
 - (i) Evidence of sufficient enrollment in any ongoing clinical trial(s) needed for marketing approval;

- (ii) Evidence of adequate progress in the development of the drug for marketing approval; and
 - (iii) Information submitted under the general investigational plan (§ 312.23(a)(3)(iv)) specifying the drug development milestones the sponsor plans to meet in the next year.
- (3) The authorization to charge is limited to the number of patients authorized to receive the drug under the treatment use, if there is a limitation.
- (4) Unless FDA specifies a shorter period, charging for expanded access to an investigational drug for treatment use under subpart I of this part may continue for 1 year from the time of FDA authorization. A sponsor may request that FDA reauthorize charging for additional periods.
- (d) *Costs recoverable when charging for an investigational drug.*
- (1) A sponsor may recover only the direct costs of making its investigational drug available.
- (i) Direct costs are costs incurred by a sponsor that can be specifically and exclusively attributed to providing the drug for the investigational use for which FDA has authorized cost recovery. Direct costs include costs per unit to manufacture the drug or costs to acquire the drug from another manufacturing source, and direct costs to ship and handle the drug.
 - (ii) Indirect costs include costs incurred primarily to produce the drug for commercial sale and research and development, administrative, labor, or other costs that would be incurred even if the clinical trial or treatment use for which charging is authorized did not occur.
- (2) For expanded access to an investigational drug for treatment use under §§ 312.315 and 312.320, in addition to the direct costs described in paragraph (d)(1)(i) of this section, a sponsor may recover the costs of monitoring the expanded access IND or protocol, complying with IND reporting requirements, and other administrative costs directly associated with the expanded access IND.
- (3) To support its calculation for cost recovery, a sponsor must provide supporting documentation to show that the calculation is consistent with the requirements of paragraphs (d)(1) and, if applicable, (d)(2) of this section. The documentation must be accompanied by a statement that an independent certified public accountant has reviewed and approved the calculations.

1.6.1 Physical AI Adaptation of § 312.8

- (e) *Physical AI system costs.* When an investigational drug is administered through or monitored by a Physical AI system, the following cost considerations apply:
- (i) The direct costs of Physical AI system deployment, including robotic hardware calibration, simulation validation, and digital twin configuration specific to the investigational drug, are not recoverable as drug costs under paragraph (d) of this section unless the Physical AI system is integral to the drug's mechanism of delivery and cannot be separated from the drug administration process.
 - (ii) Costs associated with Physical AI system safety validation, including pre-procedure safety matrix verification, USL assessment, and cross-framework simulation drug testing, are considered investigation infrastructure costs and are not recoverable as direct drug costs.
 - (iii) When a Physical AI system is classified as a combination product with the investigational drug, the sponsor may seek FDA guidance on the allocation of recoverable costs between the

drug component and the Physical AI system component.

[74 FR 40899, Aug. 13, 2009. Physical AI adaptation added 17 March 2026.]

1.7 § 312.10 Waivers

(a) A sponsor may request FDA to waive applicable requirement under this part. A waiver request may be submitted either in an IND or in an information amendment to an IND. In an emergency, a request may be made by telephone or other rapid communication means. A waiver request is required to contain at least one of the following:

- (1) An explanation why the sponsor's compliance with the requirement is unnecessary or cannot be achieved;
- (2) A description of an alternative submission or course of action that satisfies the purpose of the requirement; or
- (3) Other information justifying a waiver.

(b) FDA may grant a waiver if it finds that the sponsor's noncompliance would not pose a significant and unreasonable risk to human subjects of the investigation and that one of the following is met:

- (1) The sponsor's compliance with the requirement is unnecessary for the agency to evaluate the application, or compliance cannot be achieved;
- (2) The sponsor's proposed alternative satisfies the requirement; or
- (3) The applicant's submission otherwise justifies a waiver.

1.7.1 Physical AI Adaptation of § 312.10

(c) *Physical AI waiver provisions.* The waiver provisions of paragraphs (a) and (b) of this section apply to Physical AI system requirements established under this part. A sponsor seeking a waiver of Physical AI-specific requirements must, in addition to the requirements of paragraph (a):

- (i) Demonstrate that the waiver would not compromise the safety of human subjects who interact with or are treated by Physical AI systems during the investigation;
- (ii) Provide a risk assessment addressing the specific Physical AI system risks that the waived requirement was designed to mitigate; and
- (iii) Describe alternative safety measures that will be implemented in lieu of the waived requirement.

(d) *USL threshold waivers.* FDA may grant a waiver of the minimum USL threshold requirements specified in § 312.1(e) if the sponsor demonstrates that the Physical AI system, despite not meeting the minimum USL score, has been validated through alternative means that provide equivalent assurance of safety and effectiveness for the specific clinical procedures to be performed. Such alternative validation must be documented and submitted with the waiver request.

(e) *Simulation framework waivers.* FDA may grant a waiver of the cross-framework validation requirement if the sponsor demonstrates that the Physical AI system has been validated in a single simulation framework with sufficient fidelity and that cross-framework validation is not technically feasible for the specific system or procedure. The sponsor must provide evidence that the single-framework validation adequately characterizes the sim-to-real gap for the intended clinical use.

[52 FR 8831, Mar. 19, 1987, as amended at 52 FR 23031, June 17, 1987; 67 FR 9585, Mar. 4, 2002. Physical AI adaptation added 17 March 2026.]

2 SUBPART B — INVESTIGATIONAL NEW DRUG APPLICATION (IND)

2.1 § 312.20 Requirement for an IND

- (a) A sponsor shall submit an IND to FDA if the sponsor intends to conduct a clinical investigation with an investigational new drug that is subject to § 312.2(a).
- (b) A sponsor shall not begin a clinical investigation subject to § 312.2(a) until the investigation is subject to an IND which is in effect in accordance with § 312.40.
- (c) A sponsor shall submit a separate IND for any clinical investigation involving an exception from informed consent under § 50.24 of this chapter. Such a clinical investigation is not permitted to proceed without the prior written authorization from FDA. FDA shall provide a written determination 30 days after FDA receives the IND or earlier.

2.1.1 Physical AI Adaptation of § 312.20

- (d) When a clinical investigation involves a Physical AI system that administers, dispenses, monitors, or otherwise participates in the investigational drug delivery pathway, the sponsor shall include in the IND a Physical AI System Description as specified in § 312.23. The Physical AI System Description shall be submitted as part of the initial IND or, if the Physical AI system is introduced after the IND is in effect, as a protocol amendment under § 312.30.
- (e) A sponsor shall submit a separate protocol amendment for each distinct Physical AI system type (surgical robot, cobot, humanoid, therapeutic, diagnostic, assistive, or rehabilitative) to be used in the clinical investigation, unless the systems share substantially similar safety profiles and operational parameters, in which case they may be covered under a single protocol amendment with individual system specifications appended.
- (f) When a Physical AI system is used in an investigation involving an exception from informed consent under § 50.24 of this chapter, the IND must additionally document the Physical AI system's emergency deployment protocols, including rapid activation procedures, pre-loaded safety parameters, and the minimum human oversight requirements during emergency administration of the investigational drug.

[52 FR 8831, Mar. 19, 1987, as amended at 61 FR 51529, Oct. 2, 1996; 62 FR 32479, June 16, 1997. Physical AI adaptation added 17 March 2026.]

2.2 § 312.21 Phases of an Investigation

An IND may be submitted for one or more phases of an investigation. The clinical investigation of a previously untested drug is generally divided into three phases. Although in general the phases are conducted sequentially, they may overlap. These three phases of an investigation are as follows:

(a) *Phase 1.*

- (1) Phase 1 includes the initial introduction of an investigational new drug into humans. Phase 1 studies are typically closely monitored and may be conducted in patients or normal volunteer subjects. These studies are designed to determine the metabolism and pharmacologic actions of the drug in humans, the side effects associated with increasing doses, and, if possible, to gain early evidence on effectiveness. During Phase 1, sufficient information about the drug's

pharmacokinetics and pharmacological effects should be obtained to permit the design of well-controlled, scientifically valid, Phase 2 studies. The total number of subjects and patients included in Phase 1 studies varies with the drug, but is generally in the range of 20 to 80.

(2) Phase 1 studies also include studies of drug metabolism, structure-activity relationships, and mechanism of action in humans, as well as studies in which investigational drugs are used as research tools to explore biological phenomena or disease processes.

(b) *Phase 2.* Phase 2 includes the controlled clinical studies conducted to evaluate the effectiveness of the drug for a particular indication or indications in patients with the disease or condition under study and to determine the common short-term side effects and risks associated with the drug. Phase 2 studies are typically well controlled, closely monitored, and conducted in a relatively small number of patients, usually involving no more than several hundred subjects.

(c) *Phase 3.* Phase 3 studies are expanded controlled and uncontrolled trials. They are performed after preliminary evidence suggesting effectiveness of the drug has been obtained, and are intended to gather the additional information about effectiveness and safety that is needed to evaluate the overall benefit-risk relationship of the drug and to provide an adequate basis for physician labeling. Phase 3 studies usually include from several hundred to several thousand subjects.

2.2.1 Physical AI Adaptation of § 312.21

(d) *Phase 0 — Simulation Validation.* For clinical investigations involving Physical AI systems, a simulation validation phase (Phase 0) shall precede human subject enrollment. Phase 0 consists of comprehensive testing of the Physical AI system in validated simulation environments using patient-derived digital twin models. Phase 0 is not a regulatory phase requiring separate IND submission but shall be documented in the IND as part of the pharmacology and toxicology information under § 312.23(a)(8). Phase 0 validation shall include:

- (i) Testing in at least two recognized simulation frameworks (NVIDIA Isaac Lab v2.3.1, MuJoCo v3.4.0, Gazebo Sim v10.0.0, or PyBullet v3.2.5) demonstrating cross-framework behavioral consistency with trajectory deviations below 2mm and force discrepancies below 0.5N;
- (ii) Sim-to-real gap quantification using the validation benchmark suite with documented metrics for physics accuracy, contact dynamics, and task completion rates;
- (iii) Digital twin-based procedure rehearsal using patient-specific anatomical models calibrated to clinical imaging data; and
- (iv) Failure mode analysis across at least 1,000 simulated procedures documenting emergency stop activation rates, collision events, and force limit exceedances.

(e) *Physical AI considerations by phase.*

- (i) *Phase 1 Physical AI requirements.* When Physical AI systems are used in Phase 1 investigations, the sponsor shall implement maximum human oversight with qualified operators maintaining direct manual override capability at all times. Physical AI systems used in Phase 1 shall operate in teleoperation or supervised autonomous mode only. Fully autonomous operation is not permitted in Phase 1.
- (ii) *Phase 2 Physical AI requirements.* Phase 2 investigations may permit graduated autonomy of Physical AI systems based on Phase 1 safety data. The protocol shall specify the autonomy

level permitted and the conditions under which autonomy may be increased or decreased. USL scores shall be reassessed at the Phase 1 to Phase 2 transition using updated clinical performance data.

- (iii) *Phase 3 Physical AI requirements.* Phase 3 investigations may permit the full range of Physical AI autonomy levels supported by the accumulated safety data from Phases 1 and 2. Multi-site deployment of Physical AI systems in Phase 3 shall comply with the federated learning and cross-site coordination requirements of this part. Each site shall independently verify the Physical AI system's USL score and pre-procedure safety matrix before patient enrollment begins.

[52 FR 8831, Mar. 19, 1987. Physical AI adaptation added 17 March 2026.]

2.3 § 312.22 General Principles of the IND Submission

(a) FDA's primary objectives in reviewing an IND are, in all phases of the investigation, to assure the safety and rights of subjects, and, in Phase 2 and 3, to help assure that the quality of the scientific evaluation of drugs is adequate to permit an evaluation of the drug's effectiveness and safety. Therefore, although FDA's review of Phase 1 submissions will focus on assessing the safety of Phase 1 investigations, FDA's review of Phases 2 and 3 submissions will also include an assessment of the scientific quality of the clinical investigations and the likelihood that the investigations will yield data capable of meeting statutory standards for marketing approval.

(b) The amount of information on a particular drug that must be submitted in an IND to assure the accomplishment of the objectives described in paragraph (a) of this section depends upon such factors as the novelty of the drug, the extent to which it has been studied previously, the known or suspected risks, and the developmental phase of the drug.

(c) The central focus of the initial IND submission should be on the general investigational plan and the protocols for specific human studies. Subsequent amendments to the IND that contain new or revised protocols should build logically on previous submissions and should be supported by additional information, including the results of animal toxicology studies or other human studies as appropriate. Annual reports to the IND should serve as the focus for reporting the status of studies being conducted under the IND and should update the general investigational plan for the coming year.

(d) The IND format set forth in § 312.23 should be followed routinely by sponsors in the interest of fostering an efficient review of applications. Sponsors are expected to exercise considerable discretion, however, regarding the content of information submitted in each section, depending upon the kind of drug being studied and the nature of the available information. § 312.23 outlines the information needed for a commercially sponsored IND for a new molecular entity. A sponsor-investigator who uses, as a research tool, an investigational new drug that is already subject to a manufacturer's IND or marketing application should follow the same general format, but ordinarily may, if authorized by the manufacturer, refer to the manufacturer's IND or marketing application in providing the technical information supporting the proposed clinical investigation. A sponsor-investigator who uses an investigational drug not subject to a manufacturer's IND or marketing application is ordinarily required to submit all technical information supporting the IND, unless such information may be referenced from the scientific literature.

2.3.1 Physical AI Adaptation of § 312.22

(e) *Physical AI submission principles.* In addition to the factors listed in paragraph (b) of this section, the amount of Physical AI system information required in an IND depends upon the complexity of the Physical AI system, the degree of autonomy involved, the criticality of the system's role in drug administration, and the maturity of the system as measured by its USL score. Systems with USL scores in the Advanced band (7.0 to 10.0) may require less extensive simulation validation documentation than systems in the Foundational band (1.0 to 4.9), provided the higher USL score is supported by documented clinical deployment history.

(f) *Scalable documentation.* The Physical AI documentation requirements of § 312.23 are scalable based on the role of the Physical AI system in the investigation:

- (i) *Primary role* (the Physical AI system directly administers or dispenses the investigational drug): Full Physical AI System Description, simulation validation data, digital twin specifications, and cybersecurity assessment required.
- (ii) *Secondary role* (the Physical AI system monitors or assists with drug administration under direct human control): Abbreviated Physical AI System Description focusing on safety parameters, human-robot interaction protocols, and monitoring accuracy.
- (iii) *Ancillary role* (the Physical AI system performs supportive tasks such as sample handling, logistics, or patient positioning that do not directly involve drug contact): Summary Physical AI System Description with safety parameters and human oversight protocols.

[52 FR 8831, Mar. 19, 1987. Physical AI adaptation added 17 March 2026.]

2.4 § 312.23 IND Content and Format

(a) A sponsor who intends to conduct a clinical investigation subject to this part shall submit an "Investigational New Drug Application" (IND) including, in the following order:

- (1) *Cover sheet (Form FDA-1571).* A cover sheet for the application containing the following:
 - (i) The name, address, and telephone number of the sponsor, the date of the application, and the name of the investigational new drug.
 - (ii) Identification of the phase or phases of the clinical investigation to be conducted.
 - (iii) A commitment not to begin clinical investigations until an IND covering the investigations is in effect.
 - (iv) A commitment that an Institutional Review Board (IRB) that complies with the requirements set forth in part 56 will be responsible for the initial and continuing review and approval of each of the studies in the proposed clinical investigation and that the investigator will report to the IRB proposed changes in the research activity in accordance with the requirements of part 56.
 - (v) A commitment to conduct the investigation in accordance with all other applicable regulatory requirements.
 - (vi) The name and title of the person responsible for monitoring the conduct and progress of the clinical investigations.

- (vii) The name(s) and title(s) of the person(s) responsible under § 312.32 for review and evaluation of information relevant to the safety of the drug.
 - (viii) If a sponsor has transferred any obligations for the conduct of any clinical study to a contract research organization, a statement containing the name and address of the contract research organization, identification of the clinical study, and a listing of the obligations transferred.
 - (ix) The signature of the sponsor or the sponsor's authorized representative.
- (2) *A table of contents.*
- (3) *Introductory statement and general investigational plan.*
- (i) A brief introductory statement giving the name of the drug and all active ingredients, the drug's pharmacological class, the structural formula of the drug (if known), the formulation of the dosage form(s) to be used, the route of administration, and the broad objectives and planned duration of the proposed clinical investigation(s).
 - (ii) A brief summary of previous human experience with the drug, with reference to other IND's if pertinent, and to investigational or marketing experience in other countries that may be relevant to the safety of the proposed clinical investigation(s).
 - (iii) If the drug has been withdrawn from investigation or marketing in any country for any reason related to safety or effectiveness, identification of the country(ies) where the drug was withdrawn and the reasons for the withdrawal.
 - (iv) A brief description of the overall plan for investigating the drug product for the following year, including: (a) the rationale for the drug or the research study; (b) the indication(s) to be studied; (c) the general approach to be followed in evaluating the drug; (d) the kinds of clinical trials to be conducted in the first year following the submission; (e) the estimated number of patients to be given the drug in those studies; and (f) any risks of particular severity or seriousness anticipated on the basis of the toxicological data in animals or prior studies in humans with the drug or related drugs.
- (4) [Reserved]
- (5) *Investigator's brochure.* If required under § 312.55, a copy of the investigator's brochure, containing the following information:
- (i) A brief description of the drug substance and the formulation, including the structural formula, if known.
 - (ii) A summary of the pharmacological and toxicological effects of the drug in animals and, to the extent known, in humans.
 - (iii) A summary of the pharmacokinetics and biological disposition of the drug in animals and, if known, in humans.
 - (iv) A summary of information relating to safety and effectiveness in humans obtained from prior clinical studies.

- (v) A description of possible risks and side effects to be anticipated on the basis of prior experience with the drug under investigation or with related drugs, and of precautions or special monitoring to be done as part of the investigational use of the drug.
- (6) *Protocols.*
- (i) A protocol for each planned study. In general, protocols for Phase 1 studies may be less detailed and more flexible than protocols for Phase 2 and 3 studies. Phase 1 protocols should be directed primarily at providing an outline of the investigation—an estimate of the number of patients to be involved, a description of safety exclusions, and a description of the dosing plan including duration, dose, or method to be used in determining dose—and should specify in detail only those elements of the study that are critical to safety, such as necessary monitoring of vital signs and blood chemistries. Modifications of the experimental design of Phase 1 studies that do not affect critical safety assessments are required to be reported to FDA only in the annual report.
 - (ii) In Phases 2 and 3, detailed protocols describing all aspects of the study should be submitted.
 - (iii) A protocol is required to contain the following: (a) a statement of the objectives and purpose of the study; (b) the name, address, and qualifications of each investigator and subinvestigator, the research facilities, and the reviewing IRB; (c) the criteria for patient selection and exclusion and an estimate of the number of patients; (d) a description of the design of the study, including the control group and methods to minimize bias; (e) the method for determining the dose(s), the planned maximum dosage, and the duration of individual patient exposure; (f) a description of the observations and measurements to fulfill the study objectives; and (g) a description of clinical procedures, laboratory tests, or other measures to monitor the effects of the drug and to minimize risk.
- (7) *Chemistry, manufacturing, and control information.*
- (i) As appropriate for the particular investigations covered by the IND, a section describing the composition, manufacture, and control of the drug substance and the drug product. The amount of information needed will vary with the phase of the investigation, the proposed duration, the dosage form, and the amount of information otherwise available.
 - (ii) The amount of information to be submitted depends upon the scope of the proposed clinical investigation.
 - (iii) As drug development proceeds and as the scale of production is changed, the sponsor should submit information amendments to supplement the initial information.
 - (iv) Based on the phase(s) to be studied, the submission is required to contain: (a) *Drug substance*: a description including physical, chemical, or biological characteristics; manufacturer name and address; general method of preparation; acceptable limits and analytical methods; and stability information. (b) *Drug product*: a list of all components, quantitative composition, manufacturer name and address, manufacturing and packaging procedure, acceptable limits and analytical methods, and stability information. (c) A brief general description of the composition, manufacture, and control of any placebo. (d) *Labeling*: a copy of all labels and labeling to be provided to each investigator. (e) *Environmental analysis requirements*: a claim for categorical exclusion under § 25.30 or 25.31 or an environmental assessment under

§ 25.40.

- (8) *Pharmacology and toxicology information.* Adequate information about pharmacological and toxicological studies of the drug involving laboratory animals or in vitro, on the basis of which the sponsor has concluded that it is reasonably safe to conduct the proposed clinical investigations. Such information is required to include the identification and qualifications of the individuals who evaluated the results and a statement of where the investigations were conducted and where the records are available for inspection.
- (i) *Pharmacology and drug disposition.* A section describing the pharmacological effects and mechanism(s) of action of the drug in animals, and information on the absorption, distribution, metabolism, and excretion of the drug, if known.
- (ii) *Toxicology.* (a) An integrated summary of the toxicological effects of the drug in animals and in vitro, including acute, subacute, and chronic toxicity tests; tests of effects on reproduction and the developing fetus; and any special toxicity tests. (b) For each toxicology study primarily supporting the safety of the proposed clinical investigation, a full tabulation of data suitable for detailed review.
- (iii) For each nonclinical laboratory study subject to good laboratory practice regulations under part 58, a statement of compliance or a brief statement of the reason for noncompliance.
- (9) *Previous human experience with the investigational drug.* A summary of previous human experience known to the applicant, if any, with the investigational drug, including:
- (i) Detailed information about previous investigations or marketing experience relevant to the safety of the proposed investigation.
- (ii) If the drug is a combination of drugs previously investigated or marketed, the required information for each active drug component.
- (iii) If the drug has been marketed outside the United States, a list of the countries in which it has been marketed and withdrawn from marketing for safety or effectiveness reasons.
- (10) *Additional information.*
- (i) *Drug dependence and abuse potential.* If the drug is a psychotropic substance or otherwise has abuse potential, relevant clinical studies and experience and studies in test animals.
- (ii) *Radioactive drugs.* If the drug is a radioactive drug, sufficient data for dosimetry calculations.
- (iii) *Pediatric studies.* Plans for assessing pediatric safety and effectiveness.
- (iv) *Other information.* A brief statement of any other information that would aid evaluation of the proposed clinical investigations.
- (11) *Relevant information.* If requested by FDA, any other relevant information needed for review of the application.
- (b) *Information previously submitted.* The sponsor ordinarily is not required to resubmit information previously submitted, but may incorporate the information by reference. A reference to information submitted previously must identify the file by name, reference number, volume, and page number.

- (c) *Material in a foreign language.* The sponsor shall submit an accurate and complete English translation of each part of the IND that is not in English.
- (d) *Number of copies.* The sponsor shall submit an original and two copies of all submissions to the IND file.
- (e) *Numbering of IND submissions.* Each submission relating to an IND is required to be numbered serially using a single, three-digit serial number. The initial IND is required to be numbered 000.
- (f) *Identification of exception from informed consent.* If the investigation involves an exception from informed consent under § 50.24 of this chapter, the sponsor shall prominently identify on the cover sheet that the investigation is subject to the requirements in § 50.24 of this chapter.

2.4.1 Physical AI Adaptation of § 312.23

- (g) *Physical AI System Description.* When the clinical investigation involves a Physical AI system, the IND shall include, as a separate section following the information required by paragraph (a)(8), a Physical AI System Description containing the following:
 - (i) *System identification and classification.* The manufacturer, model designation, software version, and robot category (surgical, cobot, humanoid, therapeutic, diagnostic, assistive, or rehabilitative) of each Physical AI system to be used. The current USL score for each system, computed across the four dimensions (simulation switching, AI integration, cross-robot sharing, clinical trial collaboration), with supporting documentation for the score calculation.
 - (ii) *Role in drug administration.* A detailed description of the Physical AI system's role in the investigational drug pathway, classified as primary (direct drug administration or dispensing), secondary (monitoring or assisted administration under human control), or ancillary (supportive tasks not involving direct drug contact). The description shall include the specific clinical procedures to be performed, the degree of autonomy permitted, and the human oversight requirements for each procedure.
 - (iii) *Hardware specifications.* Technical specifications of the robotic platform, including degrees of freedom, payload capacity, positional accuracy, force sensing resolution, workspace dimensions, and end-effector specifications relevant to drug administration. For surgical robots, the specifications shall include instrument tip accuracy, tremor compensation parameters, and tissue interaction force limits.
 - (iv) *Software and AI/ML specifications.* A description of all software components, including the operating system, middleware (ROS 2 Jazzy or Kilted where applicable), control algorithms, and embedded AI/ML models. For each AI/ML model, the description shall include the model architecture, training data characteristics (including demographic diversity and oncology-specific representation), validation methodology, performance metrics, and the Predetermined Change Control Plan (PCCP) if the model is expected to be updated during the investigation.
 - (v) *Simulation validation data.* The results of Phase 0 simulation validation as described in § 312.21(d), including cross-framework behavioral consistency metrics, sim-to-real gap quantification, failure mode analysis results, and digital twin procedure rehearsal outcomes. The validation data shall reference the specific simulation framework versions used and the validation benchmark suite (v1.0) metrics achieved.

- (vi) *Digital twin specifications.* If digital twins are used for treatment planning or intraoperative guidance, a description of the digital twin modeling methodology, calibration data sources, update frequency, validation metrics, and the mathematical frameworks employed (reaction-diffusion equations, pharmacokinetic/pharmacodynamic models, finite element analysis). The description shall include the digital twin's role in the drug administration decision pathway.
 - (vii) *Safety systems.* A comprehensive description of the Physical AI system's safety architecture, including emergency stop mechanisms (hardware and software), force and torque limits, collision detection and avoidance systems, workspace boundary enforcement, and fail-safe behaviors. The description shall include the pre-procedure safety matrix requirements and the automated verification checks performed before each clinical procedure.
 - (viii) *Cybersecurity assessment.* An assessment of the Physical AI system's cybersecurity posture, including network architecture, authentication and authorization mechanisms (deny-by-default per the national MCP-PAI standard), encryption protocols, vulnerability management procedures, and incident response plans. The assessment shall address threats specific to robotic systems in clinical settings, including unauthorized command injection, sensor data manipulation, and communication interception.
 - (ix) *MCP integration.* If the Physical AI system interfaces with clinical data through the Model Context Protocol, a description of the MCP server topology, the tools and resources accessed, the conformance level achieved (Core, Clinical Read, Imaging, Federated Site, or Robot Procedure), and the hash-chained audit trail implementation. The description shall include the robot capability profile (JSON schema) and the task order lifecycle management approach.
 - (x) *Human-robot interaction protocols.* A description of the interfaces between the Physical AI system and clinical personnel, including operator control stations, status displays, alarm systems, communication channels, and handoff procedures. The description shall specify the training requirements for personnel who operate or supervise the Physical AI system during the investigation.
 - (xi) *Maintenance and calibration.* The scheduled maintenance intervals, calibration procedures, and performance verification protocols for the Physical AI system. The description shall include the criteria for taking a system out of service and the re-qualification procedures required before returning the system to clinical use.
- (h) *Investigator's brochure Physical AI supplement.* When Physical AI systems are used in the investigation, the investigator's brochure required under paragraph (a)(5) shall include a Physical AI supplement containing:
- (i) A plain-language summary of the Physical AI system's capabilities and limitations for investigator reference;
 - (ii) Known and anticipated risks specific to Physical AI system interactions, including robotic malfunction modes, AI prediction error rates, and human-robot interaction hazards;
 - (iii) Emergency procedures for Physical AI system failures, including manual override instructions and patient stabilization protocols; and
 - (iv) The minimum training requirements and competency assessments for investigators and subinvestigators who will operate or supervise the Physical AI system.

(i) *Protocol Physical AI section.* When Physical AI systems are used in the investigation, each protocol submitted under paragraph (a)(6) shall include a Physical AI section containing:

- (i) The specific Physical AI system(s) to be used in the study, referenced by the system identification in the Physical AI System Description;
- (ii) The autonomy level permitted for each procedure type and the conditions under which autonomy may be increased, decreased, or revoked;
- (iii) The Physical AI-specific inclusion and exclusion criteria for subjects (e.g., anatomical constraints, implant compatibility, claustrophobia assessment for enclosed robotic systems);
- (iv) The Physical AI-specific observations and measurements, including system telemetry data, force profiles, trajectory accuracy, and procedure timing;
- (v) The Physical AI-specific stopping rules, including system performance degradation thresholds, consecutive failure limits, and sim-to-real divergence criteria; and
- (vi) The plan for federated learning coordination if the study involves multi-site deployment of Physical AI systems, including data harmonization protocols, differential privacy parameters, and aggregation algorithms.

[52 FR 8831, Mar. 19, 1987, as amended at 52 FR 23031, June 17, 1987; 53 FR 1918, Jan. 25, 1988; 61 FR 51529, Oct. 2, 1996; 62 FR 40599, July 29, 1997; 63 FR 66669, Dec. 2, 1998; 65 FR 56479, Sept. 19, 2000; 67 FR 9585, Mar. 4, 2002. Physical AI adaptation added 17 March 2026.]

2.5 § 312.30 Protocol Amendments

Once an IND is in effect, a sponsor shall amend it as needed to ensure that the clinical investigations are conducted according to protocols included in the application. This section sets forth the provisions under which new protocols may be submitted and changes in previously submitted protocols may be made. Whenever a sponsor intends to conduct a clinical investigation with an exception from informed consent for emergency research as set forth in § 50.24 of this chapter, the sponsor shall submit a separate IND for such investigation.

(a) *New protocol.* Whenever a sponsor intends to conduct a study that is not covered by a protocol already contained in the IND, the sponsor shall submit to FDA a protocol amendment containing the protocol for the study. Such study may begin provided two conditions are met:

- (1) The sponsor has submitted the protocol to FDA for its review; and
- (2) the protocol has been approved by the Institutional Review Board (IRB) with responsibility for review and approval of the study in accordance with the requirements of part 56. The sponsor may comply with these two conditions in either order.

(b) *Changes in a protocol.*

- (1) A sponsor shall submit a protocol amendment describing any change in a Phase 1 protocol that significantly affects the safety of subjects or any change in a Phase 2 or 3 protocol that significantly affects the safety of subjects, the scope of the investigation, or the scientific quality of the study. Examples of changes requiring an amendment under this paragraph include:

- (i) Any increase in drug dosage or duration of exposure of individual subjects to the drug beyond that in the current protocol, or any significant increase in the number of subjects under study.
 - (ii) Any significant change in the design of a protocol (such as the addition or dropping of a control group).
 - (iii) The addition of a new test or procedure that is intended to improve monitoring for, or reduce the risk of, a side effect or adverse event; or the dropping of a test intended to monitor safety.
- (2) A protocol change under paragraph (b)(1) of this section may be made provided the sponsor has submitted the change to FDA and the change has been approved by the IRB. Notwithstanding the foregoing, a protocol change intended to eliminate an apparent immediate hazard to subjects may be implemented immediately provided FDA is subsequently notified by protocol amendment and the reviewing IRB is notified in accordance with § 56.104(c).
- (c) *New investigator.* A sponsor shall submit a protocol amendment when a new investigator is added to carry out a previously submitted protocol, except that a protocol amendment is not required when a licensed practitioner is added in the case of a treatment protocol under § 312.315 or § 312.320. The sponsor shall notify FDA of the new investigator within 30 days.
- (d) *Content and format.* A protocol amendment is required to be prominently identified as such and to contain the following:
- (1) In the case of a new protocol, a copy of the new protocol and a brief description of the most clinically significant differences between it and previous protocols. In the case of a change in protocol, a brief description of the change. In the case of a new investigator, the investigator's name, qualifications, reference to the previously submitted protocol, and all additional required information.
 - (2) Reference to specific technical information in the IND or in a concurrently submitted information amendment that the sponsor relies on to support any clinically significant change.
 - (3) If the sponsor desires FDA to comment on the submission, a request for such comment and the specific questions FDA's response should address.
- (e) *When submitted.* A sponsor shall submit a protocol amendment for a new protocol or a change in protocol before its implementation. Protocol amendments to add a new investigator or to provide additional information about investigators may be grouped and submitted at 30-day intervals.

2.5.1 Physical AI Adaptation of § 312.30

- (f) *Physical AI protocol amendments.* In addition to the changes described in paragraph (b)(1) of this section, the following Physical AI system changes require a protocol amendment:
- (i) Introduction of a new Physical AI system type or model not previously described in the Physical AI System Description under § 312.23(g);
 - (ii) Any change in the autonomy level permitted for a Physical AI system during clinical procedures, including changes from teleoperation to supervised autonomy or from supervised autonomy to full autonomy;

- (iii) Any significant update to an AI/ML model embedded in a Physical AI system that is outside the scope of the Predetermined Change Control Plan (PCCP), including changes to model architecture, training data composition, or performance thresholds;
 - (iv) Any change in the simulation framework or validation methodology used for ongoing Physical AI system verification;
 - (v) Any change in the human oversight requirements for Physical AI system operation, including changes to operator-to-system ratios, monitoring protocols, or intervention thresholds;
 - (vi) Any change in the digital twin modeling methodology, calibration data sources, or validation criteria that affects treatment planning or intraoperative guidance decisions; and
 - (vii) Any change in the MCP server topology, conformance level, or security architecture that affects the Physical AI system's access to clinical data.
- (g) *AI/ML model updates within PCCP scope.* Changes to AI/ML models embedded in Physical AI systems that fall within the scope of a previously submitted PCCP do not require a protocol amendment, provided the sponsor maintains documentation of all model updates and includes a summary of PCCP-covered changes in the annual report under § 312.33. The sponsor shall notify FDA within 15 calendar days if a PCCP-covered update results in a performance metric falling below the pre-specified acceptance criteria.
- (h) *Emergency Physical AI modifications.* A Physical AI system modification intended to eliminate an apparent immediate safety hazard may be implemented immediately, consistent with paragraph (b)(2) of this section. The sponsor shall submit a protocol amendment describing the emergency modification within 15 working days and shall include an assessment of the modification's impact on system safety, USL score, and ongoing simulation validation.

[52 FR 8831, Mar. 19, 1987, as amended at 52 FR 23031, June 17, 1987; 53 FR 1918, Jan. 25, 1988; 61 FR 51530, Oct. 2, 1996; 67 FR 9585, Mar. 4, 2002; 74 FR 40942, Aug. 13, 2009. Physical AI adaptation added 17 March 2026.]

2.6 § 312.31 Information Amendments

- (a) *Requirement for information amendment.* A sponsor shall report in an information amendment essential information on the IND that is not within the scope of a protocol amendment, IND safety reports, or annual report. Examples of information requiring an information amendment include:
- (1) New toxicology, chemistry, or other technical information; or
 - (2) A report regarding the discontinuance of a clinical investigation.
- (b) *Content and format of an information amendment.* An information amendment is required to bear prominent identification of its contents and to contain: (1) a statement of the nature and purpose of the amendment; (2) an organized submission of the data in a format appropriate for scientific review; and (3) if the sponsor desires FDA to comment, a request for such comment.
- (c) *When submitted.* Information amendments to the IND should be submitted as necessary but, to the extent feasible, not more than every 30 days.

2.6.1 Physical AI Adaptation of § 312.31

(d) *Physical AI information amendments.* In addition to the examples in paragraph (a), the following Physical AI-related information requires an information amendment:

- (i) New simulation validation data, including results from updated simulation framework versions or new cross-framework validation studies;
- (ii) Updated USL scores for Physical AI systems used in the investigation, whether resulting from new assessment data or changes to the USL scoring methodology;
- (iii) New cybersecurity vulnerability information affecting Physical AI systems used in the investigation, including vendor security advisories and patch deployment status;
- (iv) Changes to the national MCP-PAI standard or the Physical AI system's MCP conformance level that affect the system's interaction with clinical data;
- (v) New published literature or regulatory actions in other jurisdictions related to the safety or performance of Physical AI systems of the same type used in the investigation; and
- (vi) Updates to the digital twin modeling framework, including new calibration data sources, algorithm improvements, or validation results that are not significant enough to require a protocol amendment.

[52 FR 8831, Mar. 19, 1987, as amended at 52 FR 23031, June 17, 1987; 53 FR 1918, Jan. 25, 1988; 67 FR 9585, Mar. 4, 2002. Physical AI adaptation added 17 March 2026.]

2.7 § 312.32 IND Safety Reporting

(a) *Definitions.* The following definitions of terms apply to this section:

Adverse event means any untoward medical occurrence associated with the use of a drug in humans, whether or not considered drug related.

Life-threatening adverse event or life-threatening suspected adverse reaction. An adverse event or suspected adverse reaction is considered “life-threatening” if, in the view of either the investigator or sponsor, its occurrence places the patient or subject at immediate risk of death. It does not include an adverse event or suspected adverse reaction that, had it occurred in a more severe form, might have caused death.

Serious adverse event or serious suspected adverse reaction. An adverse event or suspected adverse reaction is considered “serious” if, in the view of either the investigator or sponsor, it results in any of the following outcomes: death, a life-threatening adverse event, inpatient hospitalization or prolongation of existing hospitalization, a persistent or significant incapacity or substantial disruption of the ability to conduct normal life functions, or a congenital anomaly/birth defect. Important medical events that may not result in death, be life-threatening, or require hospitalization may be considered serious when, based upon appropriate medical judgment, they may jeopardize the patient or subject and may require medical or surgical intervention to prevent one of the listed outcomes.

Suspected adverse reaction means any adverse event for which there is a reasonable possibility that the drug caused the adverse event. For the purposes of IND safety reporting, “reasonable possibility” means there is evidence to suggest a causal relationship between the drug and the adverse event.

Unexpected adverse event or unexpected suspected adverse reaction. An adverse event or suspected adverse reaction is considered “unexpected” if it is not listed in the investigator brochure or is not listed at the specificity or severity that has been observed; or, if an investigator brochure is not required or available, is not consistent with the risk information described in the general investigational plan or elsewhere in the current application, as amended.

(b) *Review of safety information.* The sponsor must promptly review all information relevant to the safety of the drug obtained or otherwise received by the sponsor from foreign or domestic sources, including information derived from any clinical or epidemiological investigations, animal or in vitro studies, reports in the scientific literature, and unpublished scientific papers, as well as reports from foreign regulatory authorities and reports of foreign commercial marketing experience for drugs that are not marketed in the United States.

(c) *IND safety reports.*

(1) The sponsor must notify FDA and all participating investigators in an IND safety report of potential serious risks, from clinical trials or any other source, as soon as possible, but in no case later than 15 calendar days after the sponsor determines that the information qualifies for reporting under paragraph (c)(1)(i), (c)(1)(ii), (c)(1)(iii), or (c)(1)(iv) of this section. In each IND safety report, the sponsor must identify all IND safety reports previously submitted concerning a similar suspected adverse reaction, and must analyze the significance of the suspected adverse reaction in light of previous reports or any other relevant information.

(i) *Serious and unexpected suspected adverse reaction.* The sponsor must report any suspected adverse reaction that is both serious and unexpected, where there is evidence to suggest a causal relationship between the drug and the adverse event.

(ii) *Findings from other studies.* The sponsor must report any findings from epidemiological studies, pooled analysis, or clinical studies that suggest a significant risk in humans exposed to the drug.

(iii) *Findings from animal or in vitro testing.* The sponsor must report any findings from animal or in vitro testing that suggest a significant risk in humans exposed to the drug, such as reports of mutagenicity, teratogenicity, or carcinogenicity.

(iv) *Increased rate of occurrence of serious suspected adverse reactions.* The sponsor must report any clinically important increase in the rate of a serious suspected adverse reaction over that listed in the protocol or investigator brochure.

(v) *Submission of IND safety reports.* The sponsor must submit each IND safety report in a narrative format or on FDA Form 3500A or in an electronic format that FDA can process, review, and archive.

(2) *Unexpected fatal or life-threatening suspected adverse reaction reports.* The sponsor must also notify FDA of any unexpected fatal or life-threatening suspected adverse reaction as soon as possible but in no case later than 7 calendar days after the sponsor’s initial receipt of the information.

(3) *Reporting format or frequency.* FDA may require a sponsor to submit IND safety reports in a format or at a frequency different than that required under this paragraph.

- (4) *Investigations of marketed drugs.* A sponsor of a clinical study of a drug marketed or approved in the United States that is conducted under an IND is required to submit IND safety reports for suspected adverse reactions observed in the clinical study.
- (5) *Reporting study endpoints.* Study endpoints (e.g., mortality or major morbidity) must be reported to FDA as described in the protocol and ordinarily would not be reported under paragraph (c) of this section, unless a serious and unexpected adverse event occurs for which there is evidence suggesting a causal relationship between the drug and the event.
- (d) *Followup.*
 - (1) The sponsor must promptly investigate all safety information it receives.
 - (2) Relevant followup information to an IND safety report must be submitted as soon as the information is available and must be identified as a “Followup IND Safety Report.”
 - (3) If the results of a sponsor’s investigation show that an adverse event not initially determined to be reportable under paragraph (c) of this section is so reportable, the sponsor must report it as soon as possible, but in no case later than 15 calendar days after the determination is made.
- (e) *Disclaimer.* A safety report or other information submitted by a sponsor under this part does not necessarily reflect a conclusion by the sponsor or FDA that the report or information constitutes an admission that the drug caused or contributed to an adverse event.

2.7.1 Physical AI Adaptation of § 312.32

- (f) *Physical AI safety event definitions.* In addition to the definitions in paragraph (a), the following definitions apply to safety reporting for investigations involving Physical AI systems:

Physical AI adverse event means any untoward occurrence associated with the operation of a Physical AI system during a clinical investigation involving an investigational drug, whether or not considered to be caused by the Physical AI system. Physical AI adverse events include robotic malfunction, AI prediction error, sensor failure, communication loss, unauthorized system access, digital twin divergence, and any event requiring activation of the emergency stop.

Serious Physical AI adverse event means a Physical AI adverse event that results in, or has the potential to result in, any of the outcomes listed in the definition of serious adverse event in paragraph (a), or that results in unplanned interruption of an investigational drug administration procedure requiring clinical intervention, uncontrolled release of the investigational drug, or compromise of the Physical AI system’s sterile field during a surgical procedure.

- (g) *Physical AI safety reporting requirements.* The sponsor must include Physical AI adverse events in IND safety reports as follows:

- (i) *Serious Physical AI adverse events.* Any serious Physical AI adverse event must be reported under the same timelines and procedures as paragraph (c)(1) of this section. The report must include the Physical AI system identification, the nature of the malfunction or error, the system telemetry data for the period surrounding the event, the emergency stop activation status, and the patient outcome.
- (ii) *Physical AI system-drug interaction events.* Any event in which a Physical AI system error directly or indirectly results in incorrect drug administration (wrong dose, wrong route, wrong

- timing, wrong patient) must be reported as an IND safety report regardless of whether the event meets the criteria for a serious adverse event.
- (iii) *Cybersecurity incidents.* Any cybersecurity incident affecting a Physical AI system during a clinical investigation, including unauthorized access attempts, data integrity violations, and communication interceptions, must be reported in an IND safety report within 15 calendar days of detection. If the incident results in or has the potential to result in patient harm, the 7-calendar-day reporting timeline for unexpected fatal or life-threatening events applies.
 - (iv) *AI model performance degradation.* The sponsor must report any sustained degradation in AI/ML model performance below the pre-specified acceptance criteria documented in the PCCP or the Physical AI System Description. Sustained degradation is defined as performance below acceptance criteria for three or more consecutive procedures or for a cumulative period of 24 hours, whichever occurs first.
 - (v) *Sim-to-real divergence.* The sponsor must report any instance in which the observed behavior of a Physical AI system during a clinical procedure diverges from the simulated behavior by more than twice the validated sim-to-real gap tolerance (i.e., trajectory deviation greater than 4mm or force discrepancy greater than 1.0N).
 - (vi) *Digital twin discrepancy.* The sponsor must report any instance in which a digital twin model used for treatment planning or intraoperative guidance produces predictions that diverge from observed clinical outcomes by more than the pre-specified accuracy thresholds documented in the protocol.
- (h) *Physical AI safety review.* In addition to the review requirements of paragraph (b), the sponsor must establish a Physical AI Safety Review Committee (or equivalent function) responsible for the ongoing review and evaluation of Physical AI adverse events. The committee shall include members with expertise in robotic systems engineering, AI/ML, clinical oncology, and cybersecurity. The committee shall meet at intervals not exceeding 90 days during active clinical investigations to review Physical AI safety data and to recommend modifications to Physical AI system operation, human oversight requirements, or USL threshold criteria.
- (i) *Hash-chained audit trail preservation.* For all Physical AI adverse events reported under this section, the sponsor must preserve the complete hash-chained audit trail for the Physical AI system covering the period from 24 hours before the event to 72 hours after the event. The audit trail must be maintained in accordance with the electronic records requirements of 21 CFR Part 11 and must be made available to FDA upon request.

[75 FR 59961, Sept. 29, 2010. Physical AI adaptation added 17 March 2026.]

2.8 § 312.33 Annual Reports

A sponsor shall within 60 days of the anniversary date that the IND went into effect, submit a brief report of the progress of the investigation that includes:

- (a) *Individual study information.* A brief summary of the status of each study in progress and each study completed during the previous year, including: (1) the title of the study, its purpose, a brief statement identifying the patient population, and whether the study is completed; (2) the total number of subjects initially planned, the number entered to date tabulated by age group, gender, and race, the number whose participation was completed as planned, and the number who dropped

out for any reason; and (3) if the study has been completed or interim results are known, a brief description of any available study results.

(b) *Summary information.* Information obtained during the previous year's clinical and nonclinical investigations, including: (1) a narrative or tabular summary of the most frequent and most serious adverse experiences by body system; (2) a summary of all IND safety reports submitted during the past year; (3) a list of subjects who died during participation in the investigation, with the cause of death for each; (4) a list of subjects who dropped out in association with any adverse experience; (5) a brief description of pertinent findings regarding the drug's actions; (6) a list of preclinical studies completed or in progress during the past year; and (7) a summary of any significant manufacturing or microbiological changes made during the past year.

(c) A description of the general investigational plan for the coming year to replace that submitted 1 year earlier.

(d) If the investigator brochure has been revised, a description of the revision and a copy of the new brochure.

(e) A description of any significant Phase 1 protocol modifications made during the previous year and not previously reported to the IND in a protocol amendment.

(f) A brief summary of significant foreign marketing developments with the drug during the past year.

(g) If desired by the sponsor, a log of any outstanding business with respect to the IND for which the sponsor requests or expects a reply, comment, or meeting.

2.8.1 Physical AI Adaptation of § 312.33

(h) *Physical AI system performance summary.* In addition to the information required by paragraphs (a) through (g), the annual report for investigations involving Physical AI systems shall include:

- (i) A summary of all Physical AI adverse events reported during the previous year, organized by system type, event category (malfunction, AI error, cybersecurity, digital twin discrepancy), and patient outcome;
- (ii) Updated USL scores for each Physical AI system used in the investigation, with a comparison to the scores reported in the initial IND submission or the most recent annual report, and an explanation of any changes;
- (iii) A summary of AI/ML model updates implemented during the year, distinguishing between updates within PCCP scope and updates requiring protocol amendments, and including performance metrics before and after each update;
- (iv) Updated simulation validation results, including any new cross-framework validation studies, updated sim-to-real gap measurements, and the results of any re-validation prompted by Physical AI system modifications;
- (v) A summary of digital twin model performance, including calibration accuracy, prediction reliability, and any instances of clinically significant divergence from observed outcomes;
- (vi) A summary of cybersecurity events affecting Physical AI systems, including vulnerability disclosures, patch deployments, and incident investigations;

- (vii) A summary of the Physical AI Safety Review Committee’s meetings, findings, and recommendations during the year; and
- (viii) An updated Physical AI investigational plan for the coming year, including planned system upgrades, new Physical AI system introductions, and changes to autonomy levels or human oversight requirements.

[52 FR 8831, Mar. 19, 1987, as amended at 52 FR 23031, June 17, 1987; 63 FR 6862, Feb. 11, 1998; 67 FR 9585, Mar. 4, 2002. Physical AI adaptation added 17 March 2026.]

2.9 § 312.38 Withdrawal of an IND

- (a) At any time a sponsor may withdraw an effective IND without prejudice.
- (b) If an IND is withdrawn, FDA shall be so notified, all clinical investigations conducted under the IND shall be ended, all current investigators notified, and all stocks of the drug returned to the sponsor or otherwise disposed of at the request of the sponsor in accordance with § 312.59.
- (c) If an IND is withdrawn because of a safety reason, the sponsor shall promptly so inform FDA, all participating investigators, and all reviewing Institutional Review Boards, together with the reasons for such withdrawal.

2.9.1 Physical AI Adaptation of § 312.38

(d) *Physical AI system decommissioning upon IND withdrawal.* When an IND is withdrawn and the investigation involved Physical AI systems, the sponsor shall, in addition to the requirements of paragraphs (a) through (c):

- (i) Ensure that all Physical AI systems are safely deactivated and removed from clinical service in accordance with the decommissioning procedures specified in the Physical AI System Description;
- (ii) Preserve all Physical AI system logs, telemetry data, hash-chained audit trails, simulation validation records, and digital twin models for the retention period specified in § 312.57;
- (iii) Securely erase all patient-specific data from Physical AI system local storage in compliance with HIPAA Safe Harbor de-identification requirements, while maintaining de-identified copies in the sponsor’s central records;
- (iv) Notify all sites of the Physical AI system decommissioning requirements and verify completion of decommissioning activities at each site within 30 days of the IND withdrawal; and
- (v) If the withdrawal is for a safety reason related to the Physical AI system, provide FDA with a detailed technical analysis of the Physical AI system failure modes and the contributing factors, within 60 days of the withdrawal.

[52 FR 8831, Mar. 19, 1987, as amended at 52 FR 23031, June 17, 1987; 67 FR 9586, Mar. 4, 2002. Physical AI adaptation added 17 March 2026.]